

iGaming Platform Data Modernization : Proposed Solution

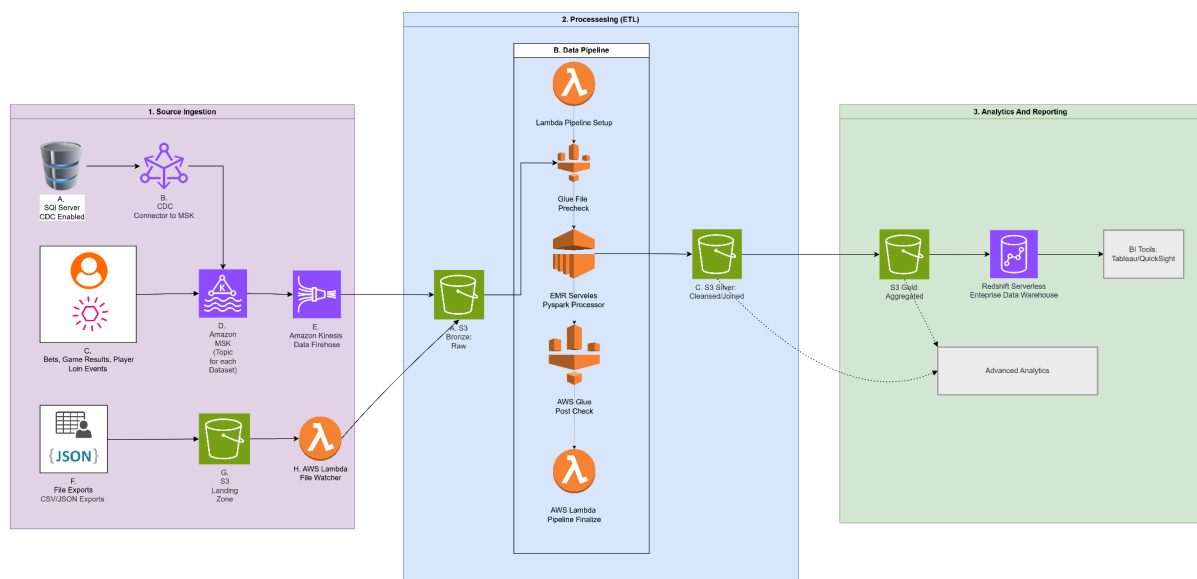
Background

BetMatrix is a rapidly growing gambling leader currently constrained by a legacy, on-premises data infrastructure that relies on daily batch processing. While the company manages a diverse ecosystem of sports betting, casino, and poker services, its existing architecture suffers from a 24-hour data latency that prevents stakeholders from executing real-time promotions or effective risk management. This bottleneck is created by the lack of integration between high-velocity Kafka event streams and the central analytics platform, as well as an over-reliance on manual, error-prone processing of third-party file exports. These technical limitations have resulted in fragmented data silos and inconsistent player metrics across departments, severely hindering the data science team's ability to develop accurate fraud detection and player behavior models.

Navigating the diagram: The solution diagram comprises 3 sections namely 1. **Source Ingestion** 2. **Processing/ETL**, 3. **Analytics and Reporting Platforms**. For the flow commentary, each component within the sections will be alphabetically numbered. For an example 1.A will be referring to the SQL server component in the source section and 2.A will be the s3 Bronze Bucket in the Ingestion Section.

Solution

To address these challenges, the proposed solution leverages AWS cloud services to move BetMatrix toward a unified, real-time data architecture designed to eliminate information silos and processing delays. By integrating live transactional data, event streams, and external provider feeds into a centralized **Datalake** on AWS, the company will achieve a single source of truth for all player metrics.



A. Streamline the ingestion and processing of data from SQL Server, Kafka, and file exports

- 1) The major step to streamlining the ingestion across all source systems we first provisioned the Amazon Managed Streaming for Apache Kafka (**Amazon MSK**) for all our data streaming capabilities as it is a fully managed service (1.D) with Amazon Kinesis Data Firehose (1.E) for write the messages to s3 Bronze Bucket (2.A) and both services (MSK and Data Firehose) can auto scale based on volume or demand.
- 2) For the Player accounts, bets, deposits/withdrawals Data coming from the Transactional Database we enabled Change Data Capture on the SQL server (1.A) in order to keep track of all the changed records that we need to process into the Datalake in near real-time. With a CDC connector to the MSK and setup topics for each dataset/tables of interest. This approach makes it easier to expand in the future should we require more data from the Transactional Database which were not part of the legacy ingestion.
- 3) Move all the topics from the Legacy Kafka instance to the newly provisioned MSK for Bets, Game Results, Player Login and Logout Events (1.C)
- 4) For the File Exports Daily exports from game providers, affiliate marketing data, and third-party verification services files to be delivered in a landing zone S3 bucket (1.G) with bucket events enabled to trigger a lambda function (1.H) to move the files to the bronze bucket as soon as it lands.

B. Scale to handle 5x data volume during major sporting events

1. On Bronze S3 Bucket (2.A) enable event to trigger pipeline (2.B) after a set time period after object creation, this can be configurable to cater for high peak seasons or events such as world cup etc.
2. Services to be set to auto scaling to accommodate unexpected increase in data volume

C. Support both BI reporting and advanced analytics for player behavior

This design eliminates data silos and provides a single source of truth as all the data is landed, cleansed and accessed from the same platform supporting both BI reporting and Advanced analytics through the Silver and Gold Layer.

D. Optimize for cost efficiency

To maximize cost efficiency, iGaming companies should leverage automated storage and specialized compute options within AWS. Implementing S3 Intelligent-Tiering can reduce costs.

Technical Questions

SQL

1. Write a SQL query to find the top 5 players by total bet amount in the past 30 days, broken down by game type.

```
WITH player_totals AS (  
    SELECT  
        player_id,  
        game_type,  
        SUM(bet_amount) AS total_bet_amount  
    FROM bet_transactions  
    WHERE bet_timestamp >= CURRENT_DATE - INTERVAL '30 days'  
    GROUP BY player_id, game_type  
)  
ranked_players AS (  
    SELECT  
        player_id,  
        game_type,  
        total_bet_amount,  
        RANK() OVER (PARTITION BY game_type ORDER BY total_bet_amount DESC) as player_position  
    FROM player_totals  
)  
SELECT *  
FROM ranked_players  
WHERE rank_position <= 5;
```

2. **Complex Aggregation with CTEs (5 points):** Write a SQL query using Common Table Expressions (CTEs) to identify players who exhibit potential problem gambling behavior by:

- o Finding players whose daily betting amount has increased by at least 50% for three consecutive days
- o Showing their average bet amount before and during this pattern
- o Sorting results by the percentage increase in betting amount

```
WITH bets_by_day AS (  
    SELECT  
        player_id,  
        CAST(bet_timestamp AS DATE) as bet_date,  
        SUM(bet_amount) as daily_total  
    FROM bet_transactions  
    GROUP BY player_id, CAST(bet_timestamp AS DATE)  
)  
bets_over_3_days AS (  
    SELECT  
        player_id,  
        bet_date,  
        daily_total,  
        LAG(daily_total, 1) OVER (PARTITION BY player_id ORDER BY bet_date) as prev_day_1,  
        LAG(daily_total, 2) OVER (PARTITION BY player_id ORDER BY bet_date) as prev_day_2,  
        LAG(daily_total, 3) OVER (PARTITION BY player_id ORDER BY bet_date) as prev_day_3  
    FROM bets_by_day  
)  
player_pattern_over_3_days AS (  
    SELECT  
        player_id,  
        bet_date,  
        daily_total,  
        prev_day_1,  
        prev_day_2,  
        prev_day_3,  
        ((daily_total - prev_day_1) / prev_day_1) * 100 as pct_inc_1,  
        ((daily_total - prev_day_2) / prev_day_2) * 100 as pct_inc_2,  
        ((daily_total - prev_day_3) / prev_day_3) * 100 as pct_inc_3  
    FROM bets_over_3_days  
    WHERE pct_inc_1 >= 50 AND pct_inc_2 >= 50 AND pct_inc_3 >= 50  
    ORDER BY pct_inc_3 DESC
```

```

SELECT player_id, bet_date as pattern_end_date
FROM bets_over_3_days
WHERE daily_total >= prev_day_1 * 1.5
  AND prev_day_1 >= prev_day_2 * 1.5
  AND prev_day_2 >= prev_day_3 * 1.5
)

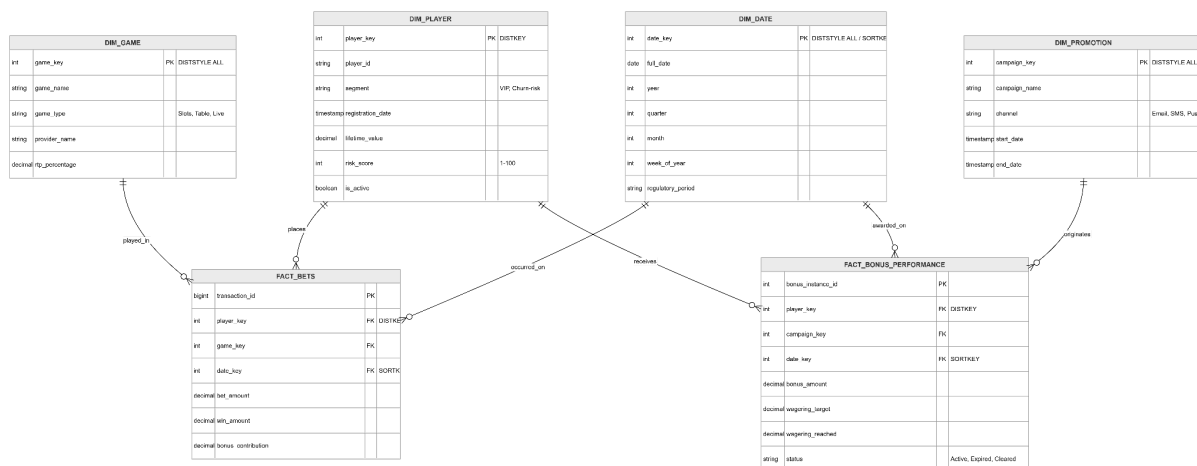
SELECT
  p.player_id,
  AVG(CASE WHEN d.bet_date BETWEEN p.pattern_end_date - INTERVAL '2 days' AND
p.pattern_end_date
      THEN d.daily_total END) as avg_during_pattern,
  AVG(CASE WHEN d.bet_date < p.pattern_end_date - INTERVAL '2 days'
      THEN d.daily_total END) as avg_before_pattern,
  ((AVG(CASE WHEN d.bet_date BETWEEN p.pattern_end_date - INTERVAL '2 days' AND
p.pattern_end_date THEN d.daily_total END) /
      NULLIF(AVG(CASE WHEN d.bet_date < p.pattern_end_date - INTERVAL '2 days' THEN
d.daily_total END), 0)) - 1) * 100 as pct_increase
FROM player_pattern_over_3_days p
JOIN bets_by_day d ON p.player_id = d.player_id
GROUP BY p.player_id
ORDER BY pct_increase DESC;

```

Data Modeling

1. Schema Design (8 points): Design a dimensional model for a cloud data warehouse (choose AWS Redshift, Azure Synapse, or Google BigQuery) that supports analysis of:

- Betting activity by game type, player segment, and time
- Bonus campaign effectiveness
- Player lifetime value and risk metrics
- Regulatory reporting requirements



Fact Tables:

fact_bets: transaction_id, player_key, game_key, date_key, bet_amount, win_amount.

Dist Key: player_key

Sort Key: date_key

fact_bonus_performance: campaign_key, player_key, bonus_id, redemption_date.

Dist Key: player_key.

Dimension Tables:

dim_players: player_key, segment, registration_date, risk_score.

Dist Style: KEY (on player_key).

dim_games: game_key, game_name, game_type.

Dist Style: ALL

dim_date: date_key, full_date, year, quarter, month, week_of_year, fin_year

Dist Style: ALL.

2. Trade-offs (7 points): Compare and contrast star schema vs. snowflake schema for this iGaming use case. How would your design accommodate both aggregated reporting and detailed player journey analysis?

Snowflake schema is storage efficient as the normalisation eliminates duplication of data, however with complex queries it introduces performance complexity, so for better analysis over big data star schema is the best choice.

3. ETL vs. ELT (5 points): Explain the differences between ETL and ELT approaches in a cloud environment. Which would you recommend for processing sensitive player data from SQL Server versus game data from file exports?

ETL stands for Extract, Transform, Load. Data is transformed after extraction before it reaches the Data Warehouse while ELT is Extract, Load, Transform which Raw data will be landed in the warehouse or lake then transformed within.

For Sensitive player data I would recommend ETL as you may need to mask or omit some sensitive data for compliance and security and for export with less sensitive data ELT is recommended.